Week-ahead CO_2 factor forecast

Summary

The forecast

- The CO_2 factor of Dutch electricity production for the next 168 hours, one value per hour, from one command.
- It is based on:
  - NED's Forecast of solar, wind, nuclear, biomass co-firing and demand for those hours
  - one LightGBM model that maps these inputs to the CO_2 factor, retrained on the 12 months of NED history before each run
- With perfect input forecasts, on the test weeks, in gCO_2/kWh:
  - average CO_2 factor: 264
  - model mean absolute error (MAE): 23, or 9% of the average
  - MAE of repeating the last available week: 111, or 42% of the average

Testing

- I chose the model on forecasts made from 25 July to 21 August 2026 and tested it on forecasts made from 7 September to 4 October 2026.
- The test uses NED's Current, its estimate of actual volumes, in place of input forecasts.

Approach

NED's CO_2 formula

- NED returns an `emissionfactor` with each of its 10 generation series, not only with the total. In the history, these factors are constant per technology, in kg/kWh:
  - gas: 0.396
  - coal: 0.798
  - CHP: 0.298
  - waste: 0.22
  - other series: 0
- This suggested that the published CO_2 factor is their mean weighted by generation:

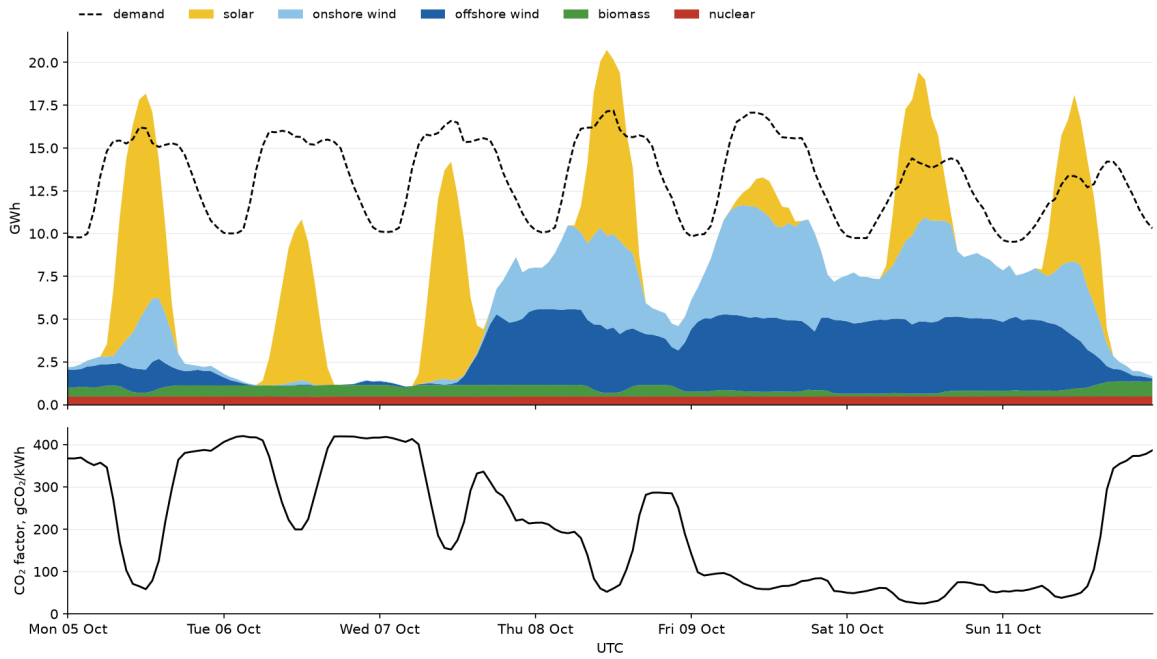
  $CO_2 \text{ factor} = 1000 \times \Sigma \text{ volume}_i \times \text{emissionfactor}_i / \text{total generation}$
- I checked it once on all 10 series, September 2024 – October 2026. The median difference from the published value is 0.02%.
- Three consequences:
  - Fossil generation cannot be a model input. Its forecasts plus this formula would give the CO_2 factor without a model, and NED publishes such a forecast itself.
  - I want a model that needs only renewable and demand forecasts, not a rebuild of NED's own CO_2 forecast. So I measure it against the weekly baseline and a linear regression, and use NED's forecast only as a sanity check.
  - The model has to learn how much demand is left for fossil plants after solar and wind. So besides renewables I add demand, residual load and the calendar.

Inputs

For each target hour, the model sees only that hour's values:

- solar, onshore and offshore wind, nuclear, biomass co-firing and demand
- residual load: demand minus solar and wind, which trees cannot easily build from four separate inputs
- the Amsterdam calendar: hour, weekday, month and a Dutch holiday flag

The CO_2 factor's own history is not an input, because the model should learn how renewable generation and demand drive the CO_2 factor, not repeat its past values.



NED's Current inputs and CO_2 factor for one week

- The chart shows the five generation inputs stacked against demand, with the CO_2 factor below, as hourly Current values for 5–11 October 2026.
- The CO_2 factor is low when the stack reaches demand and high when a gap remains for fossil plants.

Model

LightGBM with squared error loss, retrained on the 12 months before each run.

- The CO_2 factor rises with residual load nonlinearly. Gradient boosting fits this and the calendar effects.
- It trains on a year of hourly rows in seconds.
- I compared five configurations on validation and chose the most regularized one, which had the lowest MAE. Its parameters are in `co2_forecast/model.py`.
- Other training windows changed the validation error by less than 3%, so I kept 12 months.
- Linear (ridge) regression on the same inputs has about 70% higher MAE than LightGBM.

Evaluation

Protocol

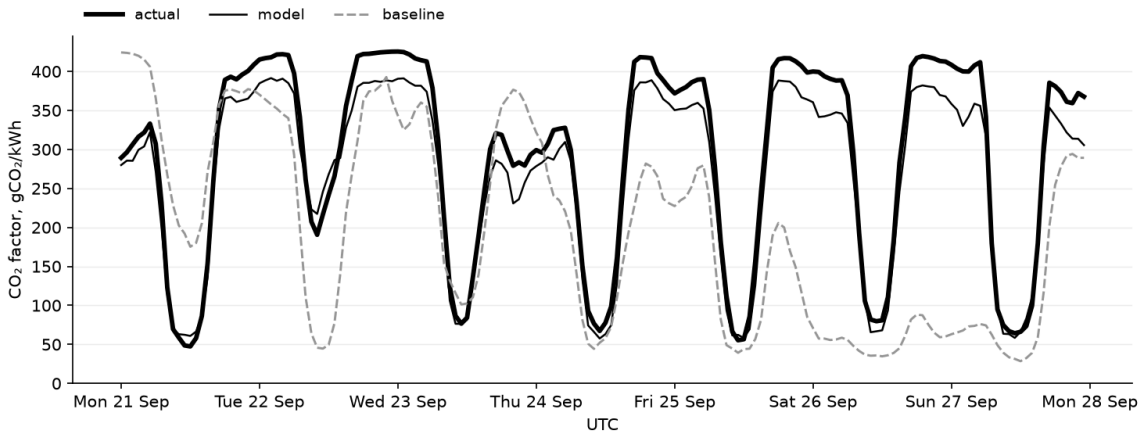
- Gaps: hours with missing NED data, such as 22–23 August 2026, are excluded from training and evaluation.
- Periods, by forecast start (origin):
  - validation: every hour from 25 July to 21 August 2026
  - test: every hour from 7 September to 4 October 2026
- Retraining: on the first day of every month, on the preceding 12 months with a one-hour availability buffer.
- Model selection: on validation only.

- Inputs: NED's Current, its estimate of actual volumes. NED's API keeps only the latest version of a past forecast, so the forecasts made at that time are not available.
- Baseline: each target hour repeats the CO_2 factor one week earlier. If that hour is not yet published when the forecast starts or is missing, it takes the last available hour before it.
- Metrics, in gCO_2/kWh, with every origin and target hour weighted equally:
  - mean absolute error (MAE): the main criterion
  - root mean squared error (RMSE): sensitive to large misses
  - mean error: forecast minus actual, negative when the model under-forecasts

Results

Metrics in gCO_2/kWh:

- Test:
  - model: MAE 23.0, RMSE 28.3, mean error −19.2
  - linear regression: MAE 38.6, RMSE 44.5, mean error −27.1
  - baseline: MAE 110.5, RMSE 139.0
- Validation:
  - model: MAE 17.8, RMSE 23.4, mean error +11.1
  - linear regression: MAE 31.7, RMSE 39.4, mean error +23.7
  - baseline: MAE 56.6, RMSE 76.0

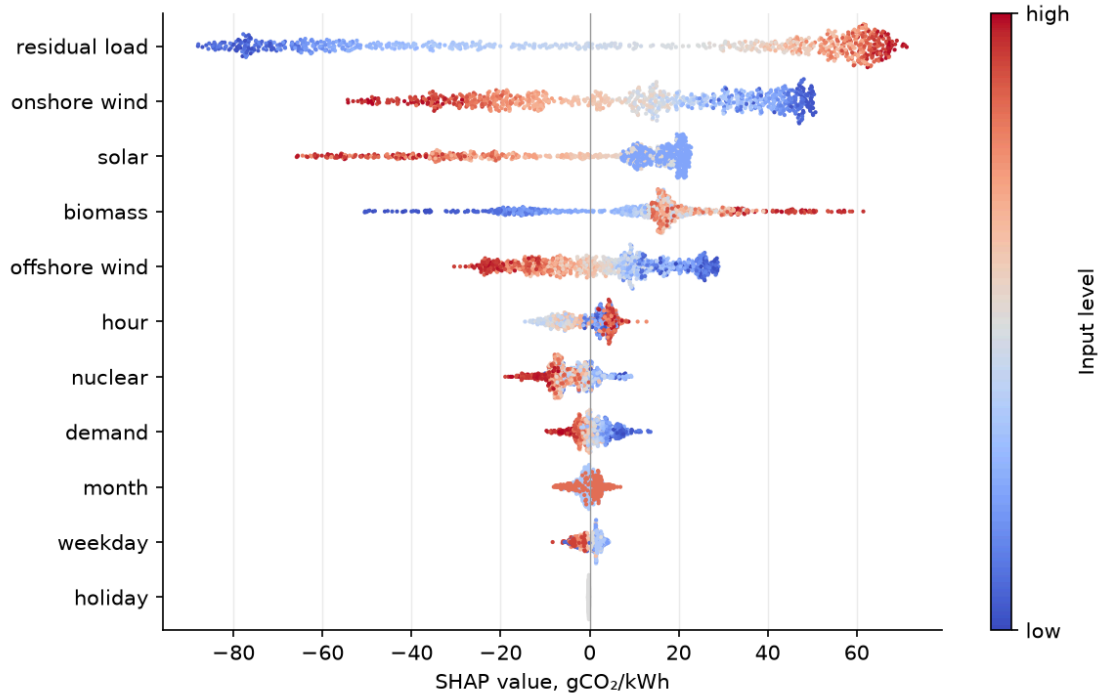


Actual, model and baseline CO_2 factor for the test week from Monday 21 September 2026

- The chart shows one test week with actual inputs.
- The model follows the daily solar dips and the multi-day wind periods that the baseline misses.
- Most of the remaining error is a level shift, as the mean error shows. The model under-forecasts on the autumn test and over-forecasts on the summer validation. The linear regression shifts the same way, so the shift does not come from boosting.
- The model does not see fuel and CO_2 prices or temperature. Prices decide whether gas or coal covers the residual load, and cold weather keeps CHP plants running for heat, so either may cause the shift.

Feature importance

- Residual load has the largest effect, followed by wind, solar and biomass. The calendar, nuclear and demand matter little.
- Residual load dominates because it measures the demand left for fossil plants.
- NED's biomass series is biomass burned in coal plants, so a high value means coal plants are running.

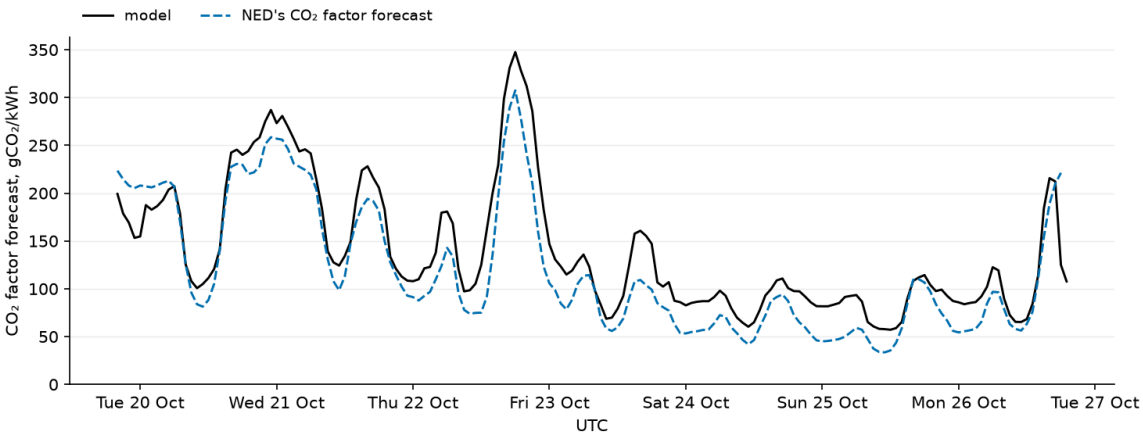


SHAP value of each input on every test hour, coloured by the input's level

- Each point is one test hour, and its colour is the input's level. High wind and solar lower the CO_2 factor, and high residual load and biomass raise it.
- Wind, solar and demand also enter residual load, so SHAP splits one effect between them. This is why high demand shows a small negative value.

Live forecast

- The run from 19 October 2026, 20:00 UTC, shows the whole pipeline on NED's live Forecast. Its shape follows NED's own CO_2 factor forecast, with correlation 0.95.

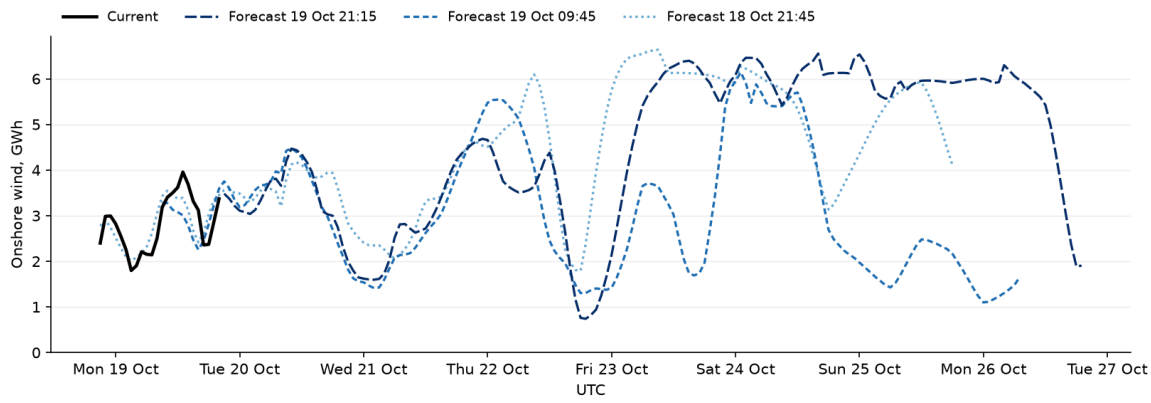


CO_2 factor forecast from 19 October 2026, 20:00 UTC, with NED's own CO_2 factor forecast

Assumptions and limitations

- The metrics assume perfect input forecasts. The real error adds the error of NED's Forecast, which grows with the horizon.
- The model learns from actual volumes and never sees forecast errors.
- The four test weeks cover one autumn. NED also revises its history, and the evaluation uses the values as downloaded.

- The last one to three forecast hours repeat the previous day, because NED's Forecast ends early or with a partial hour.
- NED revises its Forecast within hours. In two snapshots taken about a day apart, on 18–19 October 2026, the same target hours changed on average by 7–25% for wind and solar, 1% for demand and 19% for NED's own CO_2 factor forecast.
- The chart shows three onshore wind snapshots with Current for the hours already passed. They agree for the first day and diverge later.



Three snapshots of NED's onshore wind Forecast for the same target hours, with NED's Current

Next steps

- Evaluate on real forecasts. NED's API keeps only the latest Forecast, so this needs an archive of Forecast snapshots collected over a few weeks. It would give true week-ahead metrics by horizon.
- With such an archive, train on the forecasts themselves, so the model learns how much to trust them at each horizon.
- Try fuel prices, temperature and cross-border flows as inputs, to see if they fix the level shift.
- Predict quantiles instead of one value, when over- and under-forecasts cost differently.